```
!pip install -q spacy pandas
!python -m spacy download en_core_web_sm
```

```
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl (12.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 49.6 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
from google.colab import files
import pandas as pd
import spacy
df=pd.read_csv("/content/Reviews.csv")
print(df.head(5))
print(df.shape)
```

```
   Id   ProductId          UserId                      ProfileName  \
0   1  B001E4KFG0  A3SGXH7AUHU8GW                       delmartian
1   2  B00813GRG4  A1D87F6ZCVE5NK                           dll pa
2   3  B000LQOCH0   ABXLMWJIXXAIN  Natalia Corres "Natalia Corres"
3   4  B000UA0QIQ  A395BORC6FGVXV                             Karl
4   5  B006K2ZZ7K  A1UQRSCLF8GW1T    Michael D. Bigham "M. Wassir"

   HelpfulnessNumerator  HelpfulnessDenominator  Score        Time  \
0                     1                       1      5  1303862400
1                     0                       0      1  1346976000
2                     1                       1      4  1219017600
3                     3                       3      2  1307923200
4                     0                       0      5  1350777600

                Summary                                               Text
0  Good Quality Dog Food  I have bought several of the Vitality canned d...
1      Not as Advertised  Product arrived labeled as Jumbo Salted Peanut...
2  "Delight" says it all  This is a confection that has been around a fe...
3         Cough Medicine  If you are looking for the secret ingredient i...
4            Great taffy  Great taffy at a great price.  There was a wid...
(568454, 10)
```

```
reviews= df[['Text']].copy()
reviews=reviews.dropna(subset=["Text"])
reviews=reviews.drop_duplicates(subset=['Text'])
print(reviews.head(1000).reset_index(drop=True))
```

```
                                                Text
0    I have bought several of the Vitality canned d...
1    Product arrived labeled as Jumbo Salted Peanut...
2    This is a confection that has been around a fe...
3    If you are looking for the secret ingredient i...
4    Great taffy at a great price.  There was a wid...
..                                                 ...
995  Not hot at all. Like the other low star review...
996  I have to admit, I was a sucker for the large ...
```

```
997  I never in my life tasted such a good babka it...
998  I am so convinced these are human animal crack...
999  I have a whole box of peanut butter dog cookie...

[1000 rows x 1 columns]
```

```
nlp=spacy.load('en_core_web_sm')
```

```
for i, review in enumerate(reviews["Text"].head(5), start=1):
    doc = nlp(review)     print("REVIEW",
i)     print(review)     print([token.text
for token in doc])     print([token.lemma_
for token in doc])     print([token.pos_
for token in doc])     print()

REVIEW 1 I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The product looks more like a stew than a processed meat and it
smells better.
['I', 'have', 'bought', 'several', 'of', 'the', 'Vitality', 'canned', 'dog', 'food', 'products', 'and', 'have', 'found', 'them', 'all', 'to', 'be', 'of', 'good', 'quality', '.', 'The', 'pr
['I', 'have', 'buy', 'several', 'of', 'the', 'Vitality', 'can', 'dog', 'food', 'product', 'and', 'have', 'find', 'they', 'all', 'to', 'be', 'of', 'good', 'quality', '.', 'the', 'product',
['PRON', 'AUX', 'VERB', 'ADJ', 'ADP', 'DET', 'PROPN', 'VERB', 'NOUN', 'NOUN', 'NOUN', 'CCONJ', 'AUX', 'VERB', 'PRON', 'PRON', 'PART', 'AUX', 'ADP', 'ADJ', 'NOUN', 'PUNCT', 'DET', 'NOUN',
'

REVIEW 2
Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not sure if this was an error or if the vendor intended to represent the product as "Jumbo
['Product', 'arrived', 'labeled', 'as', 'Jumbo', 'Salted', 'Peanuts', '...', 'the', 'peanuts', 'were', 'actually', 'small', 'sized', 'unsalted', '.', 'Not', 'sure', 'if', 'this', 'was', 'a
['product', 'arrived', 'label', 'as', 'Jumbo', 'Salted', 'Peanuts', '...', 'the', 'peanut', 'be', 'actually', 'small', 'sized', 'unsalte', '.', 'not', 'sure', 'if', 'this', 'be', 'an', 'er
['NOUN', 'AUX', 'VERB', 'ADP', 'PROPN', 'PROPN', 'PROPN', 'PUNCT', 'DET', 'NOUN', 'AUX', 'ADV', 'ADJ', 'ADJ', 'VERB', 'PUNCT', 'PART', 'ADJ', 'SCONJ', 'PRON', 'AUX', 'DET', 'NOUN',
'CCONJ'

REVIEW 3
This is a confection that has been around a few centuries.  It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then liberally coat
['This', 'is', 'a', 'confection', 'that', 'has', 'been', 'around', 'a', 'few', 'centuries', '.', ' ', 'It', 'is', 'a', 'light', ',', 'pillowy', 'citrus', 'gelatin', 'with', 'nuts', '-', 'i
['this', 'be', 'a', 'confection', 'that', 'have', 'be', 'around', 'a', 'few', 'century', '.', ' ', 'it', 'be', 'a', 'light', ',', 'pillowy', 'citrus', 'gelatin', 'with', 'nut', '-', 'in',
['PRON', 'AUX', 'DET', 'NOUN', 'PRON', 'AUX', 'AUX', 'ADP', 'DET', 'ADJ', 'NOUN', 'PUNCT', 'SPACE', 'PRON', 'AUX', 'DET', 'ADJ', 'PUNCT', 'ADJ', 'NOUN', 'VERB', 'ADP', 'NOUN', 'PUNCT',
'AD

REVIEW 4 If you are looking for the secret ingredient in Robitussin I believe I have found it.  I got this in addition to the Root Beer Extract I ordered (which was good) and made some
cherry soda.
['If', 'you', 'are', 'looking', 'for', 'the', 'secret', 'ingredient', 'in', 'Robitussin', 'I', 'believe', 'I', 'have', 'found', 'it', '.', ' ', 'I', 'got', 'this', 'in', 'addition', 'to',
['if', 'you', 'be', 'look', 'for', 'the', 'secret', 'ingredient', 'in', 'Robitussin', 'I', 'believe', 'I', 'have', 'find', 'it', '.', ' ', 'I', 'get', 'this', 'in', 'addition', 'to', 'the'
['SCONJ', 'PRON', 'AUX', 'VERB', 'ADP', 'DET', 'ADJ', 'NOUN', 'ADP', 'PROPN', 'PRON', 'VERB', 'PRON', 'AUX', 'VERB', 'PRON', 'PUNCT', 'SPACE', 'PRON', 'VERB', 'PRON', 'ADP', 'NOUN',
'ADP',

REVIEW 5 Great taffy at a great price.  There was a wide assortment of yummy taffy.  Delivery was very quick.  If your a taffy lover, this
is a deal.
['Great', 'taffy', 'at', 'a', 'great', 'price', '.', ' ', 'There', 'was', 'a', 'wide', 'assortment', 'of', 'yummy', 'taffy', '.', ' ', 'Delivery', 'was', 'very', 'quick', '.', ' ', 'If', '
['great', 'taffy', 'at', 'a', 'great', 'price', '.', ' ', 'there', 'be', 'a', 'wide', 'assortment', 'of', 'yummy', 'taffy', '.', ' ', 'Delivery', 'be', 'very', 'quick', '.', ' ', 'if', 'yo
['ADJ', 'NOUN', 'ADP', 'DET', 'ADJ', 'NOUN', 'PUNCT', 'SPACE', 'PRON', 'VERB', 'DET', 'ADJ', 'NOUN', 'ADP', 'ADJ', 'NOUN', 'PUNCT', 'SPACE', 'PROPN', 'AUX', 'ADV', 'ADJ', 'PUNCT', 'SPACE',
```

```
doc = nlp(reviews["Text"].iloc[0]) nouns = [token.text for token
in doc if token.pos_ == "NOUN"] verbs = [token.text for token in
doc if token.pos_ == "VERB"] adjectives = [token.text for token
in doc if token.pos_ == "ADJ"] adverbs = [token.text for token in
doc if token.pos_ == "ADV"] print(nouns) print(verbs)
print(adjectives) print(adverbs)
```

```
['dog', 'food', 'products', 'quality', 'product', 'stew', 'meat', 'product']
['bought', 'canned', 'found', 'looks', 'processed', 'smells', 'appreciates']
['several', 'good', 'finicky', 'most']
['more', 'better', 'better']
```

```
from collections import Counter
pos_counter = Counter()
for doc in nlp.pipe(reviews["Text"], batch_size=16):
    for token in doc:
        if token.pos_ in ["NOUN", "VERB", "ADJ", "ADV"]:
            pos_counter[token.pos_] += 1

print(pos_counter)
```

```
Counter({'NOUN': 144487, 'VERB': 95797, 'ADJ': 71482, 'ADV': 50615})
```

```
import pandas as pd
import spacy
from collections import Counter
from google.colab import files
df = pd.read_csv("Reviews.csv")
print(df.shape)



reviews = df[["Text"]].copy()
reviews = reviews.dropna(subset=["Text"])
reviews = reviews.drop_duplicates(subset=["Text"])
reviews = reviews.head(10000).reset_index(drop=True)
print(reviews.shape)



nlp = spacy.load("en_core_web_sm")


for i, review in enumerate(reviews["Text"].head(5), start=1):
    doc = nlp(review)
    print("REVIEW", i)
    print(review)
    print([token.text for token in doc])
    print([token.lemma_ for token in doc])
    print([token.pos_ for token in doc])



doc = nlp(reviews["Text"].iloc[0])
nouns = [token.text for token in doc if token.pos_ == "NOUN"]
verbs = [token.text for token in doc if token.pos_ == "VERB"]
adjectives = [token.text for token in doc if token.pos_ == "ADJ"]
adverbs = [token.text for token in doc if token.pos_ == "ADV"]

print(nouns)
print(verbs)
```

```
print(adjectives)
print(adverbs)



pos_counter = Counter()
for doc in nlp.pipe(reviews["Text"], batch_size=100):
    for token in doc:
        if token.pos_ in ["NOUN", "VERB", "ADJ", "ADV"]:
            pos_counter[token.pos_] += 1
pos_df = pd.DataFrame(
    pos_counter.items(),
    columns=["POS Category", "Frequency"]
)
pos_df = pos_df.sort_values(
    by="Frequency",
    ascending=False
).reset_index(drop=True)
display(pos_df)



entity_data = []
for review, doc in zip(
    reviews["Text"],
    nlp.pipe(reviews["Text"], batch_size=100)
):
    for ent in doc.ents:
        if ent.label_ in [
            "PERSON",
            "ORG",
            "DATE",
            "MONEY",
            "QUANTITY",
            "PRODUCT"
        ]:
            entity_data.append({
                "Review": review,
                "Entity": ent.text,
                "Entity Label": ent.label_
            })
entity_df = pd.DataFrame(entity_data)
display(entity_df.head(20))



def extract_information(review):
    doc = nlp(review)
    return {
        "Tokens": [token.text for token in doc],
        "Lemmas": [token.lemma_ for token in doc],
        "POS Tags": [token.pos_ for token in doc],
        "Named Entities": [ent.text for ent in doc.ents],
        "Entity Labels": [ent.label_ for ent in doc.ents]
    }
```

```
new_reviews = [
    "I bought this coffee from Amazon in January 2025 for $20.",
    "John recommended this product because it has excellent taste.",
    "The package contains 5 boxes and arrived on Monday."
]
results = []
for review in new_reviews:
    result = extract_information(review)
    results.append({
        "Review": review,
        "Tokens": ", ".join(result["Tokens"]),
        "Lemmas": ", ".join(result["Lemmas"]),
        "POS Tags": ", ".join(result["POS Tags"]),
        "Named Entities": ", ".join(result["Named Entities"]),
        "Entity Labels": ", ".join(result["Entity Labels"])
    })



results_df = pd.DataFrame(results)
pd.set_option("display.max_colwidth", 200)
display(results_df)
```

```
(568454, 10)
(10000, 1)
REVIEW 1
I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The product looks more like a stew than a processed meat and it smells better. My Labrador is finicky and she appreciates this product better than  most.
['I', 'have', 'bought', 'several', 'of', 'the', 'Vitality', 'canned', 'dog', 'food', 'products', 'and', 'have', 'found', 'them', 'all', 'to', 'be', 'of', 'good', 'quality', '.', 'The', 'product', 'looks', 'more', 'like', 'a', 'stew', 'than', 'a', 'processed', 'meat', 'and', 'it', 'smells', 'better', '.', 'My', 'Labrador', 'is', 'finicky', 'and', 'she', 'appreciates', 'this', 'product', 'better', 'than', ' ', 'most', '.']
['I', 'have', 'buy', 'several', 'of', 'the', 'Vitality', 'can', 'dog', 'food', 'product', 'and', 'have', 'find', 'they', 'all', 'to', 'be', 'of', 'good', 'quality', '.', 'the', 'product', 'look', 'more', 'like', 'a', 'stew', 'than', 'a', 'process', 'meat', 'and', 'it', 'smell', 'well', '.', 'my', 'Labrador', 'be', 'finicky', 'and', 'she', 'appreciate', 'this', 'product', 'well', 'than', ' ', 'most', '.']
['PRON', 'AUX', 'VERB', 'ADJ', 'ADP', 'DET', 'PROPN', 'VERB', 'NOUN', 'NOUN', 'NOUN', 'CCONJ', 'AUX', 'VERB', 'PRON', 'PRON', 'PART', 'AUX', 'ADP', 'ADJ', 'NOUN', 'PUNCT', 'DET', 'NOUN', 'VERB', 'ADV', 'ADP', 'DET', 'NOUN', 'ADP', 'DET', 'VERB', 'NOUN', 'CCONJ', 'PRON', 'VERB', 'ADV', 'PUNCT', 'PRON', 'PROPN', 'AUX', 'ADJ', 'CCONJ', 'PRON', 'VERB', 'DET', 'NOUN', 'ADV', 'ADP', 'SPACE', 'ADJ', 'PUNCT']
REVIEW 2
Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not sure if this was an error or if the vendor intended to represent the product as "Jumbo".
['Product', 'arrived', 'labeled', 'as', 'Jumbo', 'Salted', 'Peanuts', '...', 'the', 'peanuts', 'were', 'actually', 'small', 'sized', 'unsalted', '.', 'Not', 'sure', 'if', 'this', 'was', 'an', 'error', 'or', 'if', 'the', 'vendor', 'intended', 'to', 'represent', 'the', 'product', 'as', '"', 'Jumbo', '"', '.']
['product', 'arrived', 'label', 'as', 'Jumbo', 'Salted', 'Peanuts', '...', 'the', 'peanut', 'be', 'actually', 'small', 'sized', 'unsalte', '.', 'not', 'sure', 'if', 'this', 'be', 'an', 'error', 'or', 'if', 'the', 'vendor', 'intend', 'to', 'represent', 'the', 'product', 'as', '"', 'Jumbo', '"', '.']
['NOUN', 'AUX', 'VERB', 'ADP', 'PROPN', 'PROPN', 'PROPN', 'PUNCT', 'DET', 'NOUN', 'AUX', 'ADV', 'ADJ', 'ADJ', 'VERB', 'PUNCT', 'PART', 'ADJ', 'SCONJ', 'PRON', 'AUX', 'DET', 'NOUN', 'CCONJ', 'SCONJ', 'DET', 'NOUN', 'VERB', 'PART', 'VERB', 'DET', 'NOUN', 'ADP', 'PUNCT', 'PROPN', 'PUNCT', 'PUNCT']
REVIEW 3
This is a confection that has been around a few centuries.  It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then liberally coated with powdered sugar.  And it is a tiny mouthful of heaven.  Not too chewy, and very flavorful.  I highly recommend this yummy treat.  If you are familiar with the story of C.S. Lewis' "The Lion, The Witch, and The Wardrobe" - this is the treat that seduces Edmund into selling out his Brother and Sisters to the Witch.
['This', 'is', 'a', 'confection', 'that', 'has', 'been', 'around', 'a', 'few', 'centuries', '.', ' ', 'It', 'is', 'a', 'light', ',', 'pillowy', 'citrus', 'gelatin', 'with', 'nuts', '-', 'in', 'this', 'case', 'Filberts', '.', 'And', 'it', 'is', 'cut', 'into', 'tiny', 'squares', 'and', 'then', 'liberally', 'coated', 'with', 'powdered', 'sugar', '.', ' ', 'And', 'it', 'is', 'a', 'tiny', 'mouthful', 'of', 'heaven', '.', ' ', 'Not', 'too', 'chewy', ',', 'and', 'very', 'flavorful', '.', ' ', 'I', 'highly', 'recommend', 'this', 'yummy', 'treat', '.', ' ', 'If', 'you', 'are', 'familiar', 'with', 'the', 'story', 'of', 'C.S.', 'Lewis', "'", '"', 'The', 'Lion', ',', 'The', 'Witch', ',', 'and', 'The', 'Wardrobe', '"', '-', 'this', 'is', 'the', 'treat', 'that', 'seduces', 'Edmund', 'into', 'selling', 'out', 'his', 'Brother', 'and', 'Sisters', 'to', 'the', 'Witch', '.']
['this', 'be', 'a', 'confection', 'that', 'have', 'be', 'around', 'a', 'few', 'century', '.', ' ', 'it', 'be', 'a', 'light', ',', 'pillowy', 'citrus', 'gelatin', 'with', 'nut', '-', 'in', 'this', 'case', 'filbert', '.', 'and', 'it', 'be', 'cut', 'into', 'tiny', 'square', 'and', 'then', 'liberally', 'coat', 'with', 'powdered', 'sugar', '.', ' ', 'and', 'it', 'be', 'a', 'tiny', 'mouthful', 'of', 'heaven', '.', ' ', 'not', 'too', 'chewy', ',', 'and', 'very', 'flavorful', '.', ' ', 'I', 'highly', 'recommend', 'this', 'yummy', 'treat', '.', ' ', 'if', 'you', 'be', 'familiar', 'with', 'the', 'story', 'of', 'C.S.', 'Lewis', "'", '"', 'the', 'Lion', ',', 'the', 'Witch', ',', 'and', 'the', 'Wardrobe', '"', '-', 'this', 'be', 'the', 'treat', 'that', 'seduce', 'Edmund', 'into', 'sell', 'out', 'his', 'brother', 'and', 'Sisters', 'to', 'the', 'Witch', '.']
['PRON', 'AUX', 'DET', 'NOUN', 'PRON', 'AUX', 'AUX', 'ADP', 'DET', 'ADJ', 'NOUN', 'PUNCT', 'SPACE', 'PRON', 'AUX', 'DET', 'ADJ', 'PUNCT', 'ADJ', 'NOUN', 'VERB', 'ADP', 'NOUN', 'PUNCT', 'ADP', 'DET', 'NOUN', 'NOUN', 'PUNCT', 'CCONJ', 'PRON', 'AUX', 'VERB', 'ADP', 'ADJ', 'NOUN', 'CCONJ', 'ADV', 'ADV', 'VERB', 'ADP', 'ADJ', 'NOUN', 'PUNCT', 'SPACE', 'CCONJ', 'PRON', 'AUX', 'DET', 'ADJ', 'ADJ', 'ADP', 'PROPN', 'PUNCT', 'SPACE', 'PART', 'ADV', 'NOUN', 'PUNCT', 'CCONJ', 'ADV', 'ADJ', 'PUNCT', 'SPACE', 'PRON', 'ADV', 'VERB', 'DET', 'ADJ', 'NOUN', 'PUNCT', 'SPACE', 'SCONJ', 'PRON', 'AUX', 'ADJ', 'ADP', 'DET', 'NOUN', 'ADP', 'PROPN', 'PROPN', 'PUNCT', 'PUNCT', 'DET', 'PROPN', 'PUNCT', 'DET', 'PROPN', 'PUNCT', 'CCONJ', 'DET', 'PROPN', 'PUNCT', 'PUNCT', 'PRON', 'AUX', 'DET', 'NOUN', 'PRON', 'VERB', 'PROPN', 'ADP', 'VERB', 'ADP', 'PRON', 'NOUN', 'CCONJ', 'PROPN', 'ADP', 'DET', 'PROPN', 'PUNCT']
REVIEW 4
If you are looking for the secret ingredient in Robitussin I believe I have found it.  I got this in addition to the Root Beer Extract I ordered (which was good) and made some cherry soda. The flavor is very medicinal.
['If', 'you', 'are', 'looking', 'for', 'the', 'secret', 'ingredient', 'in', 'Robitussin', 'I', 'believe', 'I', 'have', 'found', 'it', '.', ' ', 'I', 'got', 'this', 'in', 'addition', 'to', 'the', 'Root', 'Beer', 'Extract', 'I', 'ordered', '(', 'which', 'was', 'good', ')', 'and', 'made', 'some', 'cherry', 'soda', '.', ' ', 'The', 'flavor', 'is', 'very', 'medicinal', '.']
['if', 'you', 'be', 'look', 'for', 'the', 'secret', 'ingredient', 'in', 'Robitussin', 'I', 'believe', 'I', 'have', 'find', 'it', '.', ' ', 'I', 'get', 'this', 'in', 'addition', 'to', 'the', 'Root', 'Beer', 'Extract', 'I', 'order', '(', 'which', 'be', 'good', ')', 'and', 'make', 'some', 'cherry', 'soda', '.', ' ', 'the', 'flavor', 'be', 'very', 'medicinal', '.']
['SCONJ', 'PRON', 'AUX', 'VERB', 'ADP', 'DET', 'ADJ', 'NOUN', 'ADP', 'PROPN', 'PRON', 'VERB', 'PRON', 'AUX', 'VERB', 'PRON', 'PUNCT', 'SPACE', 'PRON', 'VERB', 'PRON', 'ADP', 'NOUN', 'ADP', 'DET', 'PROPN', 'PROPN', 'PROPN', 'PRON', 'VERB', 'PUNCT', 'PRON', 'AUX', 'ADJ', 'PUNCT', 'CCONJ', 'VERB', 'DET', 'NOUN', 'NOUN', 'PUNCT', 'SPACE', 'DET', 'NOUN', 'AUX', 'ADV', 'ADJ', 'PUNCT']
REVIEW 5
Great taffy at a great price.  There was a wide assortment of yummy taffy.  Delivery was very quick.  If your a taffy lover, this is a deal.
['Great', 'taffy', 'at', 'a', 'great', 'price', '.', ' ', 'There', 'was', 'a', 'wide', 'assortment', 'of', 'yummy', 'taffy', '.', ' ', 'Delivery', 'was', 'very', 'quick', '.', ' ', 'If', 'your', 'a', 'taffy', 'lover', ',', 'this', 'is', 'a', 'deal', '.']
['great', 'taffy', 'at', 'a', 'great', 'price', '.', ' ', 'there', 'be', 'a', 'wide', 'assortment', 'of', 'yummy', 'taffy', '.', ' ', 'Delivery', 'be', 'very', 'quick', '.', ' ', 'if', 'your', 'a', 'taffy', 'lover', ',', 'this', 'be', 'a', 'deal', '.']
['ADJ', 'NOUN', 'ADP', 'DET', 'ADJ', 'NOUN', 'PUNCT', 'SPACE', 'PRON', 'VERB', 'DET', 'ADJ', 'NOUN', 'ADP', 'ADJ', 'NOUN', 'PUNCT', 'SPACE', 'PROPN', 'AUX', 'ADV', 'ADJ', 'PUNCT', 'SPACE', 'SCONJ', 'PRON', 'DET', 'PROPN', 'NOUN', 'PUNCT', 'PRON', 'AUX', 'DET', 'NOUN', 'PUNCT']
['dog', 'food', 'products', 'quality', 'product', 'stew', 'meat', 'product']
```

```
['bought', 'canned', 'found', 'looks', 'processed', 'smells', 'appreciates']
['several', 'good', 'finicky', 'most']
['more', 'better', 'better']
```

	POS Category	Frequency
0	NOUN	144487
1	VERB	95797
2	ADJ	71482
3	ADV	50615

	Review	Entity	Entity Label
0	I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The product looks more like a stew than a processed meat and it smells better. My Labr...	Vitality	ORG
1	I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The product looks more like a stew than a processed meat and it smells better. My Labr...	My Labrador	PERSON
2	Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not sure if this was an error or if the vendor intended to represent the product as "Jumbo".	Jumbo Salted	ORG
3	This is a confection that has been around a few centuries. It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then liberally coated with ...	around a few centuries	DATE
4	This is a confection that has been around a few centuries. It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then liberally coated with ...	pillowy citrus gelatin	ORG
5	This is a confection that has been around a few centuries. It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then liberally coated with ...	C.S. Lewis'	ORG
6	This is a confection that has been around a few centuries. It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then liberally coated with ...	Edmund	PRODUCT
7	This is a confection that has been around a few centuries. It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then liberally coated with ...	Sisters	PERSON
8	This is a confection that has been around a few centuries. It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then liberally coated with ...	Witch	ORG
9	Great taffy at a great price. There was a wide assortment of yummy taffy. Delivery was very quick. If your a taffy lover, this is a deal.	Delivery	PERSON
10	I got a wild hair for taffy and ordered this five pound bag. The taffy was all very enjoyable with many flavors: watermelon, root beer, melon, peppermint, grape, etc. My only complaint is there wa...	five pound	QUANTITY
11	I got a wild hair for taffy and ordered this five pound bag. The taffy was all very enjoyable with many flavors: watermelon, root beer, melon, peppermint, grape, etc. My only complaint is there wa...	only two weeks	DATE
12	This saltwater taffy had great flavors and was very soft and chewy. Each candy was individually wrapped well. None of the candies were stuck together, which did happen in the expensive version, ...	Fralinger's	ORG
13	Right now I'm mostly just sprouting this so my cats can eat the grass. They love it. I rotate it around with Wheatgrass and Rye too	Wheatgrass	PERSON
14	I don't know if it's the cactus or the tequila or just the unique combination of ingredients, but the flavour of this hot sauce makes it one of a kind! We picked up a bottle once on a trip we wer...	Tequila Picante Gourmet de Inclan	ORG
15	My cats have been happily eating Felidae Platinum for more than two years. I just got a new bag and the shape of the food is different. They tried the new food when I first put it in their bowls a...	Felidae Platinum	ORG
16	My cats have been happily eating Felidae Platinum for more than two years. I just got a new bag and the shape of the food is different. They tried the new food when I first put it in their bowls a...	more than two years	DATE
17	good flavor! these came securely packed... they were fresh and delicious! i love these Twizzlers!	Twizzlers	ORG
18	The Strawberry Twizzlers are my guilty pleasure - yummy. Six pounds will be around for a while with my son and I.	The Strawberry Twizzlers	ORG
19	The Strawberry Twizzlers are my guilty pleasure - yummy. Six pounds will be around for a while with my son and I.	Six pounds	QUANTITY

	Review	Tokens	Lemmas	POS Tags	Named Entities	Entity Labels
0	I bought this coffee from Amazon in January 2025 for $20.	I, bought, this, coffee, from, Amazon, in, January, 2025, for, $, 20, .	I, buy, this, coffee, from, Amazon, in, January, 2025, for, $, 20, .	PRON, VERB, DET, NOUN, ADP, PROPN, ADP, PROPN, NUM, ADP, SYM, NUM, PUNCT	Amazon, January 2025, 20	ORG, DATE, MONEY
1	John recommended this product because it has excellent taste.	John, recommended, this, product, because, it, has, excellent, taste, .	John, recommend, this, product, because, it, have, excellent, taste, .	PROPN, VERB, DET, NOUN, SCONJ, PRON, VERB, ADJ, NOUN, PUNCT	John	PERSON
2	The package contains 5 boxes and arrived on Monday.	The, package, contains, 5, boxes, and, arrived, on, Monday, .	the, package, contain, 5, box, and, arrive, on, Monday, .	DET, NOUN, VERB, NUM, NOUN, CCONJ, VERB, ADP, PROPN, PUNCT	5, Monday	CARDINAL, DATE